

Griffin Edwards

Ph.D. Candidate in Mathematics, Georgia Institute of Technology
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Research Interests

Algebraic combinatorics, especially matroid theory and its connections with tropical geometry, algebraic geometry, and pastures.

Education

Georgia Institute of Technology, Atlanta, Georgia

August 2022–May 2027
(expected)

Ph.D. in Mathematics

Doctoral Minor in Higher Education and Tech to Teaching Certificate (in progress)

Center for Teaching and Learning

Dissertation (working title): *Dressians of Matroids through Pastures*

Advisor: Matthew Baker

Brown University, Providence, Rhode Island

September 2018–May 2022

Bachelor of Science in Mathematics, with Honors

Research and Scholarly Work

Manuscripts in preparation

- *Foundations of Elliptic Matroids*, with Matthew Baker and Noah Solomon. Manuscript in preparation.
- *Foundations of Matroids, Part 3*, with Matthew Baker and Oliver Lorscheid. Manuscript in preparation.

Software in development

- *Pastures*, a Macaulay2 package, with Justin Chen and Tianyi Zhang.
- *Rigidity*, a Macaulay2 package, with Kalina Mincheva, Daniel Irving Bernstein, and additional collaborators.

Selected research contribution

- Computationally verified the conjectured reduced-Dressian behavior of the elliptic matroids T_n for $7 \leq n \leq 26$; this computation is acknowledged in Baker–Huh–Kummer–Lorscheid, *Lorentzian Polynomials and Matroids over Triangular Hyperfields* (2025).

Teaching Experience

Instructor of Record, Georgia Institute of Technology

Fall 2026

MATH 1499: Support for Introductory Calculus

- Designed and taught a flipped, discussion-centered course for 15 students, using interactive pre-class readings, collaborative worksheets and quizzes, and student board presentations.
- Used an initial diagnostic assessment to identify common prerequisite gaps and incorporated targeted review into recurring class activities.
- Revised course materials to improve screen-reader accessibility and mathematical readability.

Lecturer's Assistant, Georgia Institute of Technology

Summer 2026

MATH 4107: Abstract Algebra I

Teaching Assistant, Georgia Institute of Technology

Fall 2024–Spring 2026

- *MATH 2603: Introduction to Discrete Mathematics* — Fall 2025 and Spring 2026
- *MATH 1552: Integral Calculus* — Spring 2025
- *MATH 1553: Introduction to Linear Algebra* — Fall 2024

Undergraduate Teaching Assistant, Brown University

2020–2021

Teaching support for Honors Linear Algebra, Real Analysis, Differential Geometry, Linear Algebra, and Advanced Classical Mechanics.

Tutor, Sheridan Center for Teaching and Learning, Brown University

September 2019–May 2021

Tutored students in mathematics, physics, and logic.

Research Mentoring

Research Mentor and General Helper, Georgia Tech Mathematics REU *Summer 2025*
Co-mentored Catie Lillja and Bora Olmez with Matthew Baker and Noah Solomon on *Algorithm for Group Law on Tropical Elliptic Curve*.

Mentor, Georgia Tech Directed Reading Program *Spring 2026*
Tutte Polynomial and Heawood Conjecture.

Presentations

Title to be announced. Tulane University, New Orleans, Louisiana. *November 18, 2026*
Tropical Elliptic Curves and Matroids. Poster presented at the ISM Discovery School: Combinatorics of Moduli of Curves, LACIM, Montréal, Canada. *May 11–15, 2026*
Computer Algebra Club/Seminar presentation, Georgia Institute of Technology. *November 6, 2025*
Algebra from Projective Geometry. Algebra Student Seminar, Georgia Institute of Technology. *November 17, 2023*
An Introductory Proof of the Compactness Theorem and Some Applications. Algebra Student Seminar, Georgia Institute of Technology. *October 28, 2022*

Awards and Fellowships

MathLab Award for Top TA Helper, Georgia Tech School of Mathematics *2026*
Recognized for exceptional student support in the department’s communal drop-in office-hour and study space.
GAANN Grant, U.S. Department of Education *January 2025–present*
C-PIES GradSTEP Grant, Georgia Tech College of Sciences *January–May 2025*
FESTA Fellowship, Georgia Institute of Technology *April 2024*
David L. Brown Fellowship Award, Georgia Institute of Technology *April 2023*
Research to Training Fellowship, Georgia Institute of Technology *August 2022*
Undergraduate Teaching and Research Award, Brown University *Summer 2020*

Conferences and Workshops

Macaulay2 Workshop, Tulane University; contributor *April 2025*
Tropical Geometry: Moduli Spaces and Matroids, Goethe University Frankfurt *October 2024*
Macaulay2, Computational Algebraic Geometry and String Theory; contributor *May 2024*
Preliminary Arizona Winter School; participant *Fall 2023*
Arizona Winter School; participant *March 2023*
Connecticut Summer School in Number Theory; participant *June 2022*

Service, Leadership, and Outreach

Volunteer and Problem Writer, High School Math Day, Georgia Institute of Technology *2025–2027*
College of Sciences Graduate Student Committee Member, Georgia Tech *2024–present*
With C-PIES, select speakers for the *Making Science Accessible Seminar Series*; founding organizer of the annual Bowling and Poster Competition, coordinating judges, awards, and publicity.
Graduate Peer Mentor Coordinator, Georgia Institute of Technology *2023–2026*
President, Graduate AMS Chapter, Georgia Institute of Technology *2023–2026*
Teacher, Emory Math Circle *2022–2026*
Graduate Student Representative, Georgia Institute of Technology *2023–2024*
President, Mathematics Department Undergraduate Group, Brown University *2020–2022*
Founder and Coordinator, Presentation Theory, Brown University *Fall 2021*
Planned and moderated mathematics seminars by undergraduates for undergraduates.
Directed Reading Program — *Rational Points on Elliptic Curves*, Brown University *Fall 2021*
Volunteer, Brown University Math Circle *Fall 2021–Spring 2022*

Technical Skills and Interests

Software and programming: Macaulay2, Python, Coq, and LaTeX.

Other interests: Board games, cooking, and juggling.

Last updated September 2026.